

Security incident report:

Apply OS Hardening Techniques

Section 1: Identify the network protocol involved in the incident

The network protocol involved in the incident is HyperText Transfer Protocol (HTTP). HTTP was used at the application layer to establish the connection with the website, request the web pages, and deliver the malicious executable file to the visitors' browsers.

Section 2: Document the incident

After the incident, multiple customers emailed yummyrecipesforme's helpdesk complaining that the company's website (yummyrecipesforme.com) had prompted them to download a file to access free recipes. After running the file, they mentioned that their computers began running more slowly. In response to this incident, the owner tried to log in to the admin panel, but the password to the admin account had been changed. So, the owner was unable to get access and reached out to the website hosting provider.

To address this incident, the cybersecurity analyst used a sandbox environment to observe the suspicious website behavior. Then, the analyst ran tcpdump to analyze the network traffic produced by interacting with the website. When the website loaded, the analyst was prompted to download an executable file to update the browser. When accepted and run, the browser redirected to a different URL (greatrecipesforme.com), which contained the malware.

After inspecting the tcpdump logs, the analyst observed that the browser initiated a DNS request for the IP address of the yummyrecipesforme.com from

the DNS server. After receiving the correct IP address, the browser initiated an HTTP request for the webpage. Thereafter, the analyst noticed a sudden change in the network traffic as the browser initiated the download of an executable file and a DNS request for a different website called greatrecipesforme.com. The network traffic was then rerouted to the new website after receiving its IP address from the DNS server.

A senior analyst confirmed that the website had been compromised and checked the source code for the website. The analyst discovered that JavaScript code had been added to prompt website visitors to download an executable file. Additionally, a script in that file redirected the visitors' browser from yummyrecipesforme.com to greatrecipesforme.com. The cybersecurity team reported that the web server was impacted by a brute-force attack because the hacker was able to guess the password easily since the admin password was still set to the default password. As a result, the attacker gained access to the system, implanted the script and changed the password.

Section 3: Recommend one remediation for brute force attacks

One remediation against brute-force attacks is to prohibit the reuse of previous passwords. Additionally, enforcing strong password complexity, a combination of letters, symbols and numbers makes passwords significantly harder for attackers to guess. Finally, implementing multi-factor authentication (MFA) adds a critical layer of defense, as it requires both a password and a time-sensitive one-time password (OTP), which brute-force attacks cannot easily bypass.

